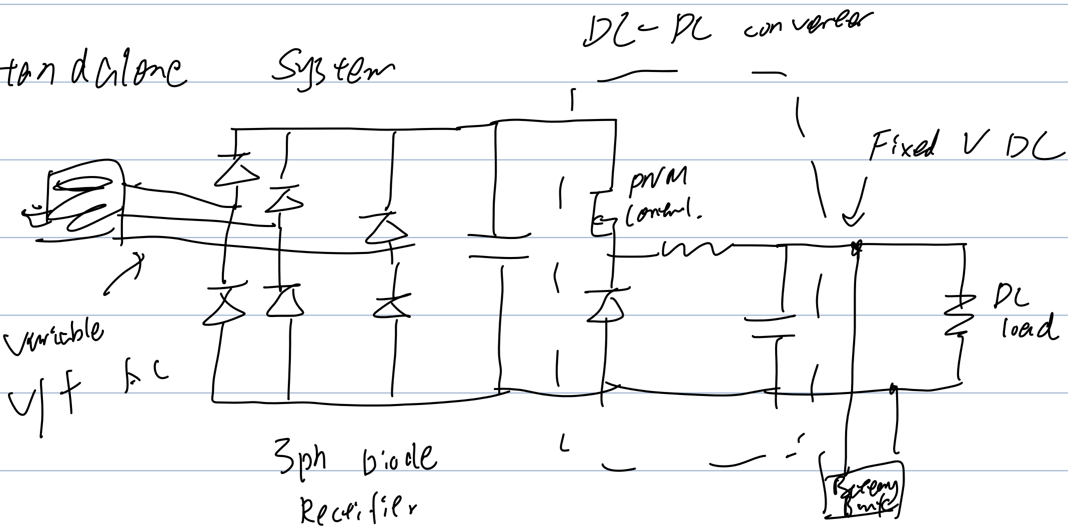
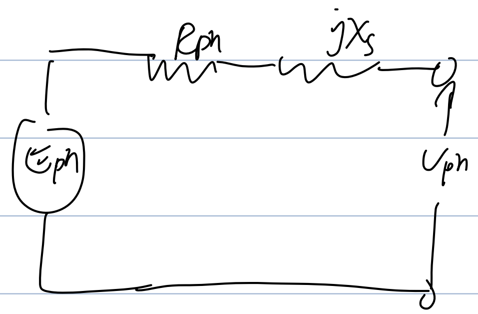
Brushless PM Gen

Standalone System





Example:

(i) $f_s = 30\ Hz$, $N_s = \frac{120 f_s}{P} = \frac{120\times 30}{10} = 360\ rpm$

$T = K_t \cdot I = 10 \cdot 6.63 = 66.3\ Nm$

$P_{mech} = T \cdot W = 66.3 \times \frac{360 \times 2\pi}{60} = 2.5\ kW$

at $P_{mech} = 2.5\ kW \Rightarrow V_o = 58\ V$

(ii) $\frac{V_{DC}^2}{R_o} = P_{mech} \Rightarrow R_o = \frac{V_{DC}^2}{P_{mech}} = \frac{12^2}{2.5\ kW} = 0.058\ \Omega$.

$D = \frac{V_{out}}{V_{in}} = 0.21$.